功能几何实验：数据与审计补充

C：混合权重平方距离；O：专家输出距离；F：模型 Fisher 二次代价；O-diag、F-diag：对应中心化 Gram 的对角控制。均值 ± SD 使用三个训练种子；配对区间按同一 checkpoint 的 1,024 个窗口做 2,000 次重采样。

协议与选择规则

四个 gate、三个种子、五种目标，共 60 项适配。每项从各自已训练的分数拟合 checkpoint 开始；同一 gate/seed 的五个目标共享全部初始参数、样本序列与固定校准幂。每项 2,000 步、batch256，学习率 0.0003；所有目标按真实内部 dev KL 从 step0 与每 200 步 checkpoint 中选择。

训练缓存来自训练集前 75%，内部 dev 来自后 25% 的独立窗口；验证集只在 checkpoint 选定后报告。训练几何、目标及 checkpoint 选择使用原模型权重和输出分布。自然标签仅用于结果报告。观察过的 validation 单列，测量范围为最终专家模块替换后的预定采样位置。

gate	实际参数	更新参数	执行路径
MLP	10385	10384	经典
Affine Born	10385	10384	经典
CZ	10313	10312	实态模拟
Product	10313	10312	实态模拟

冻结的 log_scale 仍计入总参数，适配时不更新。专家和主干参数均固定；同一 seed 共享同一模型 checkpoint。功能 Gram 每样本 16×16，输出与 Fisher 各一份；推理不读取 Gram。

数学恒等式与计量对象

$$\left\|\sum_e d_e f_e\right\|^2 = \sum_e d_e^2 \|v_e\|^2 + \sum_{e\neq j} d_e d_j \langle v_e, v_j\rangle, \quad v_e = f_e - \overline{f}$$

系数和为零时，完整输出误差对专家共同平移不变。对角能量与交叉项的拆分依赖坐标原点，本轮统一使用中心化专家。Fisher Gram 与独立 JVP、精确 Hessian 的局部关系已核验。QFI 描述状态角度；稀疏权重 Jacobian 描述固定集合内的可达权重方向。

当前目标只对被选中的权重求通常梯度；共享网络参数更新可以改变后续的选择边界，top-k 集合本身未引入替代梯度。初始 checkpoint 与适配后结果的差异同时包含追加训练与目标改变；本轮直接比较的因子为共同追加预算下的 C/O/F/对角目标。

dev：MLP / Affine Born

gate/目标	seed	最佳步	CE	PPL	ΔCE	KL
MLP/C	17	2000	4.0972	60.173	+0.00188	0.00153
MLP/C	29	1000	4.0661	58.326	+0.00091	0.00131
MLP/C	43	1800	4.0523	57.529	+0.00255	0.00137
MLP/O	17	1800	4.0970	60.157	+0.00160	0.00150
MLP/O	29	600	4.0661	58.328	+0.00093	0.00134
MLP/O	43	1800	4.0519	57.504	+0.00211	0.00138
MLP/F	17	2000	4.0964	60.121	+0.00101	0.00156
MLP/F	29	1200	4.0660	58.323	+0.00086	0.00137
MLP/F	43	1800	4.0517	57.495	+0.00196	0.00142
MLP/O-diag	17	2000	4.0968	60.148	+0.00145	0.00150
MLP/O-diag	29	1000	4.0656	58.303	+0.00051	0.00132
MLP/O-diag	43	1800	4.0519	57.507	+0.00217	0.00138
MLP/F-diag	17	2000	4.0966	60.138	+0.00130	0.00154
MLP/F-diag	29	1200	4.0661	58.328	+0.00093	0.00135
MLP/F-diag	43	1800	4.0521	57.517	+0.00233	0.00139
Affine Born/C	17	1800	4.0961	60.103	+0.00070	0.00118
Affine Born/C	29	1400	4.0659	58.315	+0.00072	0.00140
Affine Born/C	43	1800	4.0505	57.427	+0.00078	0.00097
Affine Born/O	17	2000	4.0959	60.092	+0.00052	0.00131
Affine Born/O	29	2000	4.0657	58.308	+0.00061	0.00143
Affine Born/O	43	1800	4.0509	57.449	+0.00115	0.00105
Affine Born/F	17	1600	4.0956	60.075	+0.00024	0.00127
Affine Born/F	29	2000	4.0657	58.307	+0.00058	0.00147
Affine Born/F	43	1800	4.0508	57.441	+0.00101	0.00109
Affine Born/O-diag	17	2000	4.0961	60.106	+0.00076	0.00123
Affine Born/O-diag	29	2000	4.0654	58.289	+0.00027	0.00143
Affine Born/O-diag	43	1800	4.0509	57.452	+0.00120	0.00103
Affine Born/F-diag	17	1800	4.0963	60.118	+0.00096	0.00126
Affine Born/F-diag	29	2000	4.0658	58.310	+0.00063	0.00144
Affine Born/F-diag	43	1800	4.0508	57.443	+0.00106	0.00105

ΔCE 相对各自原始模型。KL 方向为原模型到适配模型。每 seed/split 为 4,096 个位置，同一窗口内的位置共同重采样。所有候选完整列示。

dev：CZ / Product

gate/目标	seed	最佳步	CE	PPL	ΔCE	KL
CZ/C	17	2000	4.1000	60.338	+0.00460	0.00493
CZ/C	29	2000	4.0668	58.372	+0.00170	0.00219
CZ/C	43	2000	4.0565	57.771	+0.00674	0.00642
CZ/O	17	2000	4.0978	60.208	+0.00246	0.00441
CZ/O	29	2000	4.0666	58.360	+0.00149	0.00213
CZ/O	43	2000	4.0536	57.607	+0.00390	0.00562
CZ/F	17	2000	4.0987	60.259	+0.00331	0.00452
CZ/F	29	1600	4.0664	58.347	+0.00127	0.00226
CZ/F	43	1800	4.0561	57.748	+0.00634	0.00613
CZ/O-diag	17	2000	4.0986	60.254	+0.00321	0.00440
CZ/O-diag	29	2000	4.0666	58.359	+0.00147	0.00213
CZ/O-diag	43	2000	4.0542	57.639	+0.00445	0.00563
CZ/F-diag	17	2000	4.0987	60.260	+0.00332	0.00450
CZ/F-diag	29	1800	4.0662	58.335	+0.00107	0.00221
CZ/F-diag	43	2000	4.0542	57.637	+0.00443	0.00556
Product/C	17	2000	4.1022	60.471	+0.00681	0.00803
Product/C	29	2000	4.0777	59.009	+0.01255	0.01275
Product/C	43	2000	4.0592	57.929	+0.00948	0.00899
Product/O	17	1600	4.1025	60.494	+0.00719	0.00724
Product/O	29	2000	4.0750	58.851	+0.00986	0.00987
Product/O	43	1800	4.0577	57.840	+0.00795	0.00856
Product/F	17	2000	4.1034	60.547	+0.00808	0.00719
Product/F	29	2000	4.0749	58.843	+0.00972	0.00982
Product/F	43	1800	4.0585	57.890	+0.00881	0.00842
Product/O-diag	17	2000	4.1021	60.470	+0.00680	0.00726
Product/O-diag	29	2000	4.0739	58.786	+0.00876	0.01039
Product/O-diag	43	1800	4.0579	57.851	+0.00813	0.00863
Product/F-diag	17	1600	4.1027	60.506	+0.00738	0.00723
Product/F-diag	29	2000	4.0745	58.822	+0.00938	0.00955
Product/F-diag	43	1800	4.0579	57.853	+0.00816	0.00862

ΔCE 相对各自原始模型。KL 方向为原模型到适配模型。每 seed/split 为 4,096 个位置，同一窗口内的位置共同重采样。所有候选完整列示。

validation： MLP / Affine Born

gate/目标	seed	最佳步	CE	PPL	ΔCE	KL
MLP/C	17	2000	4.3462	77.185	-0.00185	0.00173
MLP/C	29	1000	4.3274	75.745	-0.00042	0.00130
MLP/C	43	1800	4.3349	76.314	+0.00073	0.00133
MLP/O	17	1800	4.3460	77.173	-0.00201	0.00161
MLP/O	29	600	4.3273	75.736	-0.00053	0.00133
MLP/O	43	1800	4.3352	76.342	+0.00109	0.00135
MLP/F	17	2000	4.3454	77.122	-0.00266	0.00172
MLP/F	29	1200	4.3268	75.702	-0.00098	0.00133
MLP/F	43	1800	4.3351	76.330	+0.00094	0.00140
MLP/O-diag	17	2000	4.3461	77.178	-0.00195	0.00164
MLP/O-diag	29	1000	4.3273	75.736	-0.00053	0.00128
MLP/O-diag	43	1800	4.3351	76.332	+0.00096	0.00136
MLP/F-diag	17	2000	4.3457	77.144	-0.00239	0.00165
MLP/F-diag	29	1200	4.3269	75.708	-0.00090	0.00131
MLP/F-diag	43	1800	4.3352	76.338	+0.00104	0.00139
Affine Born/C	17	1800	4.3461	77.178	-0.00194	0.00126
Affine Born/C	29	1400	4.3286	75.835	+0.00077	0.00141
Affine Born/C	43	1800	4.3349	76.314	+0.00072	0.00103
Affine Born/O	17	2000	4.3461	77.175	-0.00198	0.00134
Affine Born/O	29	2000	4.3282	75.809	+0.00043	0.00137
Affine Born/O	43	1800	4.3349	76.317	+0.00077	0.00109
Affine Born/F	17	1600	4.3465	77.205	-0.00159	0.00135
Affine Born/F	29	2000	4.3276	75.761	-0.00020	0.00145
Affine Born/F	43	1800	4.3347	76.301	+0.00055	0.00114
Affine Born/O-diag	17	2000	4.3464	77.199	-0.00167	0.00128
Affine Born/O-diag	29	2000	4.3283	75.815	+0.00051	0.00139
Affine Born/O-diag	43	1800	4.3348	76.307	+0.00064	0.00107
Affine Born/F-diag	17	1800	4.3468	77.229	-0.00128	0.00131
Affine Born/F-diag	29	2000	4.3281	75.801	+0.00033	0.00140
Affine Born/F-diag	43	1800	4.3352	76.340	+0.00107	0.00112

ΔCE 相对各自原始模型。KL 方向为原模型到适配模型。每 seed/split 为 4,096 个位置，同一窗口内的位置共同重采样。所有候选完整列示。

validation：CZ / Product

gate/目标	seed	最佳步	CE	PPL	ΔCE	KL
CZ/C	17	2000	4.3505	77.516	+0.00243	0.00505
CZ/C	29	2000	4.3286	75.842	+0.00086	0.00222
CZ/C	43	2000	4.3385	76.591	+0.00435	0.00619
CZ/O	17	2000	4.3519	77.628	+0.00387	0.00457
CZ/O	29	2000	4.3281	75.798	+0.00029	0.00221
CZ/O	43	2000	4.3382	76.568	+0.00405	0.00563
CZ/F	17	2000	4.3514	77.585	+0.00331	0.00478
CZ/F	29	1600	4.3282	75.811	+0.00045	0.00228
CZ/F	43	1800	4.3386	76.598	+0.00445	0.00621
CZ/O-diag	17	2000	4.3516	77.600	+0.00351	0.00454
CZ/O-diag	29	2000	4.3280	75.793	+0.00022	0.00218
CZ/O-diag	43	2000	4.3393	76.658	+0.00522	0.00563
CZ/F-diag	17	2000	4.3512	77.573	+0.00316	0.00460
CZ/F-diag	29	1800	4.3280	75.790	+0.00018	0.00220
CZ/F-diag	43	2000	4.3387	76.604	+0.00452	0.00557
Product/C	17	2000	4.3554	77.897	+0.00733	0.00839
Product/C	29	2000	4.3396	76.681	+0.01186	0.01282
Product/C	43	2000	4.3425	76.896	+0.00832	0.00857
Product/O	17	1600	4.3551	77.872	+0.00701	0.00775
Product/O	29	2000	4.3385	76.594	+0.01073	0.01023
Product/O	43	1800	4.3423	76.883	+0.00815	0.00823
Product/F	17	2000	4.3550	77.866	+0.00694	0.00760
Product/F	29	2000	4.3366	76.450	+0.00885	0.00998
Product/F	43	1800	4.3411	76.795	+0.00701	0.00800
Product/O-diag	17	2000	4.3552	77.885	+0.00717	0.00777
Product/O-diag	29	2000	4.3392	76.643	+0.01137	0.01097
Product/O-diag	43	1800	4.3424	76.891	+0.00826	0.00825
Product/F-diag	17	1600	4.3551	77.873	+0.00703	0.00769
Product/F-diag	29	2000	4.3380	76.552	+0.01018	0.00991
Product/F-diag	43	1800	4.3422	76.878	+0.00809	0.00819

ΔCE 相对各自原始模型。KL 方向为原模型到适配模型。每 seed/split 为 4,096 个位置，同一窗口内的位置共同重采样。所有候选完整列示。

跨种子汇总：validation 均值 ± SD

gate/目标	CE	PPL	KL
MLP/C	4.33615 ± 0.00949	76.415 ± 0.726	0.00145 ± 0.00024
MLP/O	4.33617 ± 0.00943	76.417 ± 0.721	0.00143 ± 0.00016
MLP/F	4.33576 ± 0.00931	76.385 ± 0.712	0.00148 ± 0.00021
MLP/O-diag	4.33615 ± 0.00947	76.415 ± 0.724	0.00143 ± 0.00019
MLP/F-diag	4.33591 ± 0.00941	76.397 ± 0.720	0.00145 ± 0.00018
Affine Born/C	4.33651 ± 0.00889	76.442 ± 0.681	0.00123 ± 0.00019
Affine Born/O	4.33640 ± 0.00902	76.434 ± 0.690	0.00127 ± 0.00015
Affine Born/F	4.33625 ± 0.00954	76.422 ± 0.730	0.00131 ± 0.00016
Affine Born/O-diag	4.33649 ± 0.00917	76.441 ± 0.702	0.00125 ± 0.00016
Affine Born/F-diag	4.33670 ± 0.00942	76.457 ± 0.721	0.00128 ± 0.00014
CZ/C	4.33921 ± 0.01094	76.650 ± 0.839	0.00449 ± 0.00204
CZ/O	4.33939 ± 0.01197	76.665 ± 0.919	0.00414 ± 0.00175
CZ/F	4.33940 ± 0.01159	76.665 ± 0.889	0.00442 ± 0.00199
CZ/O-diag	4.33964 ± 0.01178	76.683 ± 0.904	0.00412 ± 0.00176
CZ/F-diag	4.33928 ± 0.01164	76.656 ± 0.893	0.00412 ± 0.00173
Product/C	4.34583 ± 0.00839	77.158 ± 0.649	0.00993 ± 0.00251
Product/O	4.34529 ± 0.00867	77.116 ± 0.670	0.00874 ± 0.00132
Product/F	4.34426 ± 0.00957	77.037 ± 0.738	0.00852 ± 0.00127
Product/O-diag	4.34559 ± 0.00850	77.140 ± 0.657	0.00900 ± 0.00172
Product/F-diag	4.34509 ± 0.00891	77.101 ± 0.688	0.00860 ± 0.00116

SD 使用三个 seed 的样本标准差（ddof=1），与给定 checkpoint 内按窗口重采样的区间分开。PPL 为每 seed 的采样 CE 指数，表中取其均值与 SD；BPE 词表与采样协议固定。

交叉项的配对结果：完整目标减去对角目标

gate/目标	seed	dev ΔKL	dev 95% 区间	val ΔKL
MLP/O	17	+0.00001	[-0.00004, +0.00005]	-0.00003
MLP/O	29	+0.00002	[-0.00003, +0.00006]	+0.00005
MLP/O	43	+0.00000	[-0.00001, +0.00001]	-0.00001
MLP/F	17	+0.00002	[-0.00002, +0.00006]	+0.00007
MLP/F	29	+0.00002	[-0.00001, +0.00006]	+0.00001
MLP/F	43	+0.00002	[-0.00000, +0.00005]	+0.00001
Affine Born/O	17	+0.00007	[+0.00005, +0.00010]	+0.00006
Affine Born/O	29	+0.00001	[-0.00002, +0.00004]	-0.00001
Affine Born/O	43	+0.00002	[+0.00001, +0.00003]	+0.00002
Affine Born/F	17	+0.00001	[-0.00003, +0.00005]	+0.00003
Affine Born/F	29	+0.00004	[+0.00001, +0.00007]	+0.00005
Affine Born/F	43	+0.00003	[+0.00001, +0.00006]	+0.00002
CZ/O	17	+0.00001	[-0.00004, +0.00006]	+0.00002
CZ/O	29	+0.00000	[-0.00002, +0.00003]	+0.00003
CZ/O	43	-0.00001	[-0.00009, +0.00007]	+0.00001
CZ/F	17	+0.00002	[-0.00010, +0.00014]	+0.00018
CZ/F	29	+0.00005	[-0.00002, +0.00012]	+0.00008
CZ/F	43	+0.00057	[+0.00040, +0.00072]	+0.00065
Product/O	17	-0.00003	[-0.00013, +0.00008]	-0.00003
Product/O	29	-0.00052	[-0.00075, -0.00030]	-0.00073
Product/O	43	-0.00007	[-0.00014, -0.00001]	-0.00003
Product/F	17	-0.00005	[-0.00016, +0.00008]	-0.00009
Product/F	29	+0.00028	[-0.00008, +0.00070]	+0.00007
Product/F	43	-0.00020	[-0.00029, -0.00010]	-0.00019

配对 CE、完整 validation 区间及对 coefficient 的比较保存在 reviews/functional_diagonal_v1/paired.csv；原始三目标的 900 项配对统计保存在 reviews/functional_v1/paired.csv。区间反映给定 checkpoint 的窗口采样变动；训练种子差异单独保留。

数值扰动、利用率与误差机制

gate/目标	**dev BF16** 切换	有害切换	未用专家	完整集合数
MLP/C	17	1	1.3 ± 0.6	114.0 ± 33.7
MLP/O	18	0	2.0 ± 1.0	114.7 ± 24.8
MLP/F	12	0	1.7 ± 0.6	114.3 ± 24.0
MLP/O-diag	14	0	2.3 ± 0.6	110.7 ± 23.5
MLP/F-diag	18	0	1.7 ± 1.5	108.7 ± 22.5
Affine Born/C	16	0	1.7 ± 1.2	112.0 ± 20.2
Affine Born/O	16	2	1.0 ± 1.0	118.0 ± 22.9
Affine Born/F	24	1	1.3 ± 0.6	107.7 ± 23.7
Affine Born/O-diag	22	1	2.0 ± 0.0	112.7 ± 16.3
Affine Born/F-diag	10	1	2.0 ± 1.0	105.3 ± 20.5
CZ/C	30	0	0.3 ± 0.6	142.0 ± 31.1
CZ/O	17	1	0.3 ± 0.6	155.7 ± 45.2
CZ/F	21	0	0.3 ± 0.6	167.3 ± 39.9
CZ/O-diag	27	0	0.3 ± 0.6	156.3 ± 45.4
CZ/F-diag	23	2	0.3 ± 0.6	162.0 ± 40.3
Product/C	20	0	0.7 ± 1.2	48.0 ± 3.6
Product/O	24	2	0.7 ± 1.2	50.7 ± 5.0
Product/F	16	1	0.7 ± 1.2	53.7 ± 7.1
Product/O-diag	12	1	0.7 ± 1.2	48.0 ± 4.0
Product/F-diag	15	1	0.7 ± 1.2	51.0 ± 6.6

切换与有害事件为三个种子共 12,288 个位置的计数。有害定义为集合发生变化，且该次输入舍入令目标负对数概率增加超过 0.1nat。这里仅将 gate 输入做 FP16/BF16 舍入往返，计算核与专家保持 FP32；全部 FP16 记录和误差尾部见原始 result.json。

机制数据

reviews/functional_mechanism_v1 含 96 组逐位置误差分解：系数距离、原始权重遗漏质量、输出误差、中心化能量及交叉项。theory/sparse_tangent_v1 含 24 个 checkpoint 的 1,536 个位置、3,072 个有/无地板 Jacobian 与 128 个独立数学检查。前者为结果描述，后者为固定支持集的局部结构。

执行预算与串行延迟

gate/目标	**1** 行 **ms**	**256** 行 **ms**	**4096** 行 **ms**	适配秒**/run**
MLP/C	0.943 ± 0.079	0.901 ± 0.017	0.863 ± 0.010	17.9 ± 2.6
MLP/O	0.949 ± 0.087	0.930 ± 0.083	0.912 ± 0.084	18.3 ± 1.2
MLP/F	0.952 ± 0.080	0.931 ± 0.095	0.912 ± 0.099	18.3 ± 0.7
MLP/O-diag	0.890 ± 0.014	0.871 ± 0.007	0.842 ± 0.011	18.9 ± 1.8
MLP/F-diag	0.887 ± 0.014	0.886 ± 0.027	0.903 ± 0.105	17.9 ± 0.2
Affine Born/C	1.164 ± 0.004	1.156 ± 0.009	1.131 ± 0.001	19.3 ± 0.4
Affine Born/O	1.234 ± 0.120	1.180 ± 0.057	1.128 ± 0.008	19.6 ± 0.2
Affine Born/F	1.222 ± 0.085	1.160 ± 0.003	1.126 ± 0.004	20.3 ± 0.6
Affine Born/O-diag	1.214 ± 0.095	1.200 ± 0.081	1.114 ± 0.023	19.6 ± 0.4
Affine Born/F-diag	1.145 ± 0.026	1.134 ± 0.020	1.109 ± 0.016	20.9 ± 1.8
CZ/C	5.705 ± 0.022	5.745 ± 0.035	6.206 ± 0.466	58.2 ± 0.5
CZ/O	5.816 ± 0.141	5.729 ± 0.011	5.777 ± 0.055	60.0 ± 0.7
CZ/F	5.760 ± 0.049	5.751 ± 0.017	6.073 ± 0.597	59.1 ± 0.4
CZ/O-diag	5.742 ± 0.108	5.698 ± 0.059	5.964 ± 0.402	60.0 ± 1.4
CZ/F-diag	6.297 ± 0.492	5.739 ± 0.080	5.772 ± 0.038	58.8 ± 0.7
Product/C	5.889 ± 0.387	6.407 ± 0.445	6.363 ± 0.393	58.0 ± 1.1
Product/O	5.999 ± 0.487	6.447 ± 0.315	6.220 ± 0.283	59.8 ± 0.2
Product/F	5.737 ± 0.109	5.996 ± 0.536	5.728 ± 0.090	59.6 ± 1.2
Product/O-diag	5.994 ± 0.471	5.714 ± 0.018	6.089 ± 0.407	58.6 ± 0.8
Product/F-diag	6.143 ± 0.453	5.951 ± 0.397	5.925 ± 0.473	58.8 ± 0.7

计时逐 checkpoint 串行执行，20 次预热、50 次 CUDA event 重复；以上对三个种子的 median 取均值与 SD。执行时间包括 router、稀疏选择与固定校准幂。原始 50 次、P95、吞吐、增量显存、GPU 现场在 timing/functional_v1。TITAN V 与其他进程共享，测量范围为该运行环境的 router 路径。

适配秒数包括训练与该脚本内的检查，不包括共享的几何预计算。完整 36 项与对角 24 项的起止状态、预计算文件和日志均保留。对角控制复用已经验证的训练 Gram，完整列出原始来源及哈希。

完成核验与可追溯路径

工作	核验
完整功能目标	36 训练、72 质量、144 扰动、900 配对，独立 CPU 审计通过
对角控制	24 训练、48 质量、96 扰动、48CPU checkpoint 重放
适配后计时	60checkpoint、180 组、每组 50 次重复
误差分解	96 记录，Gram 恒等式与平移不变性通过
稀疏切空间	24checkpoint、3,072Jacobian；独立微分与有限差分通过
缺失/失败（本轮正式组）	空；旧选择器失败 v1、强正则 pilot 与阴性研究完整保留

主要结果目录：D:/files/MOTE/outputs/operator_geometry_round3。functional/50b8c7704256；functional_diagonal/260b32fdf440；reviews/functional_v1；reviews/functional_diagonal_v1；reviews/functional_mechanism_v1；theory/sparse_tangent_v1；timing/functional_v1。每组保存配置、冻结源码哈希、checkpoint、采样数组、训练曲线与原始 NPZ。

先前阶段报告：STAGE_REVIEW_20261030_zh 与 CAPACITY_GEOMETRY_REVIEW_20261030_zh，分别对应条件代价/几何机制与容量/稀疏重放阶段。当前报告新增功能适配与交叉项控制。阶段用时按用户授权排除网络中断；不重复旧训练填充预算。

文献与后续评价

Balanced Knowledge Distillation for Large Language Models with Mix-of-Experts

Knowledge Distillation for Mixture of Experts Models in Speech Recognition

On the Expressive Power and Capacity of Quantum Data Reuploaders

后续共同适配实验应独立预注册任务成绩与资源判据，允许不同于 teacher 的专家分工。当前已观察 validation 用于探索性汇报，未来扩大时需预留新的评估切分或任务。

先前量子路由新颖性与最近对照核查